MediScribe AI

Ambient Clinical Intelligence & Multilingual EHR Automation Copilot

"Restoring the Doctor-Patient Bond through Regional Code-Switching AI & Deterministic Pharmacology Safety"

80%	<5ms	100%	SHA-256
TIME SAVED	SAFETY LATENCY	ALLERGY SHIELD	EHR SIGN-OFF

END-TO-END AMBIENT CLINICAL PIPELINE

1. Ambient Audio
Passive doctor-patient dialogue capture (Malayalam + English)

→

2. AI Synthesis
Structured SOAP note & WHO ICD-10 diagnostic codes

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3. Safety Shield
Sub-5ms deterministic allergy & drug interaction checks

→

4. 1-Click Dispatch
Official signed prescription PDF + Malayalam care guide

CORE BREAKTHROUGH INNOVATIONS

1. Regional Code-Switching AI

- Parses colloquial terms ('[redacted]', '[redacted]') into standard medical terminology.
- Anti-hallucination: unsaid vitals are marked null, never invented.
- Auto-generates WHO ICD-10 codification.

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2. Deterministic Safety Shield

- **Allergy Cross-Reactivity:** Blocks beta-lactams & amoxicillin in penicillin-allergic patients.
- **Drug-Drug Interactions:** Flags NSAIDs with anticoagulants or ACEi + ARBs.
- **Duplicate Protection:** Prevents paracetamol hepatotoxicity.

Safety Check: [CRITICAL] Amoxicillin blocked for Penicillin Allergy

3. Bilingual Discharge & PDF

- **ReportLab TrueType Engine:** Flawless native Malayalam rendering (Noto Sans Malayalam).
- **Patient Care Guide:** Clear schedule ([redacted], [redacted], [redacted] [redacted]).
- **Digital Verification:** Doctor KMC license bound with SHA-256 hash.

മലയാളം നിർദ്ദേശം:
പാരസെറ്റാമോൾ 650
ഭക്ഷണത്തിന് ശേഷം

TECHNICAL STACK & CLINICAL IMPACT

Asynchronous Microservice Architecture:

- **Backend:** FastAPI, Python 3.12/3.14, SQLAlchemy 2.0, Pydantic v2
- **Frontend:** React 18, TypeScript, Vite, Tailwind CSS, Lucide UI
- **AI Core:** Google Gemini API with Multi-Key Round-Robin Failover
- **Clinical RAG:** ChromaDB Guideline Indexer & Vector Search
- **PDF Engine:** ReportLab 5.0 with TrueType HarfBuzz shaping
- **Resilience:** 100% Offline Simulation Mode for air-gapped clinics

Metric	Traditional OPD	MediScribe AI
Documentation Time	6–8 min / pt	**30–45 sec / pt**
Eye-Contact with Patient	<30% (Typing)	**>90% (Active)**
Allergy Catch Rate	Memory dependent	**100% Automated**
Patient Discharge Guide	English / Illegible	**Malayalam + Eng**

MediScribe AI — Outpatient Clinical Copilot
Major Project Expo 2026 | Department of Computer Science & Engineering

Live Interactive Demo
Scan for GitHub Source & Clinical Docs